CASE STUDY

Multi-Plant Production Unification

in Automotive Manufacturing

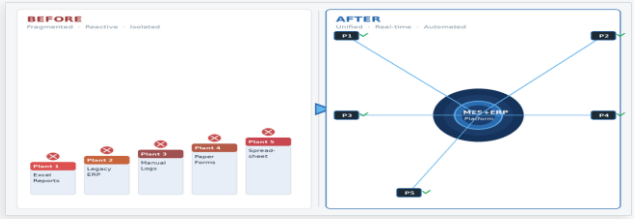
A unified digital platform integrating 5 plants for production, quality, inventory & compliance.



CS-AUTO-MPU-2024

BACKGROUND

A Tier-1 automotive supplier operating across 5 manufacturing plants in different regions faced challenges in managing production, quality, and inventory uniformly. Each plant used different systems (Excel, legacy ERP, manual logs), leading to inconsistencies.



KEY CHALLENGES

a. Data Silos

- No centralized visibility across plants
- Production, quality & inventory data stored separately

b. Inconsistent Processes

- Different SOPs, quality checks and reporting formats
- Difficult to maintain compliance with IATF 16949

c. Lack of Real-Time Monitoring

- Delayed decision-making
- No live production tracking across plants

e. Inefficient Planning

- Poor coordination between plants
- Overstock in one plant, shortages in another

OBJECTIVE

To implement a unified digital platform that integrates all plants for:

- **Production Monitoring**
- **Quality Management**
- **Inventory Synchronization**
- **Traceability & Compliance**

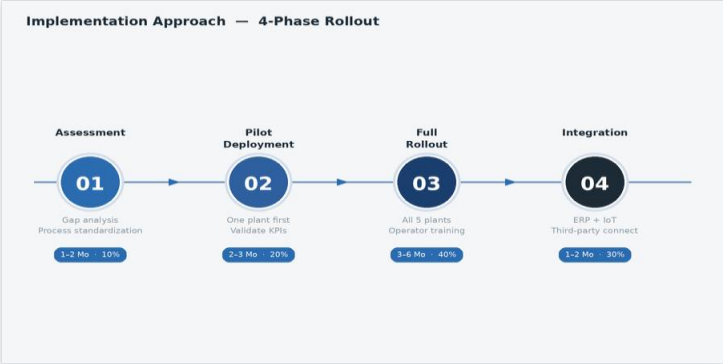
SOLUTION IMPLEMENTED — CENTRALIZED MANUFACTURING PLATFORM (MES + ERP INTEGRATION)

A cloud-based system deployed across all plants:

a. Production Management	Real-time tracking; Standardized work orders
b. Quality Management (QMS)	Integrated SPC, MSA, Control Plan, PPAP
c. Inventory & SCM	Central RM, WIP, FG; Inter-plant optimization
d. Traceability System	End-to-end batch & serial; Barcode tracking
e. Management Dashboard	Live KPIs: OEE, Rejection, Downtime, Delivery



IMPLEMENTATION APPROACH

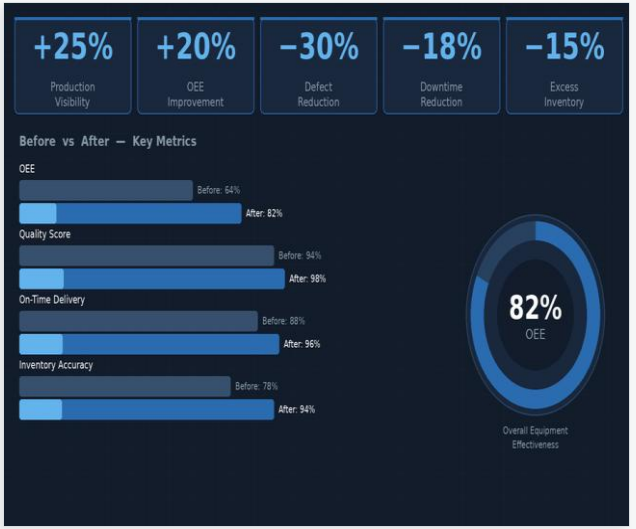


01	**Phase 1: Assessment** Gap analysis across plants • Process standardization	Timeline **1–2 Months**	Budget **10%**	Deliverable: Gap Analysis Report
02	**Phase 2: Pilot Deployment** Implemented in one plant • Validated workflows and KPIs	Timeline **2–3 Months**	Budget **20%**	Deliverable: Pilot Validation Report
03	**Phase 3: Rollout** Gradual deployment to all plants • Operator & manager training	Timeline **3–6 Months**	Budget **40%**	Deliverable: Training & Deployment Docs
04	**Phase 4: Integration** Integrated with ERP, IoT machines, and third-party systems	Timeline **1–2 Months**	Budget **30%**	Deliverable: Integration Sign-off

KEY RESULTS

+25%	+20%	−30%	−18%	−15%
Production Visibility	OEE Improvement	Defect Reduction	Downtime Reduction	Excess Inventory

Category	Improvement	Area
Production Visibility	**25% increase**	Operational
OEE	**20% improvement**	Operational
Downtime	**18% reduction**	Operational
Defect Rate	**30% reduction**	Quality
Audit Readiness	**Improved**	IATF 16949
Excess Inventory	**15% reduction**	Inventory



BEFORE vs AFTER

✕ BEFORE

P1	P2	P3	P4	P5
Excel / Manual	Legacy ERP	Paper Logs	Spreadsheets	Manual Forms

No shared connection between plants

- No centralized visibility across plants
- Different SOPs, quality checks & formats
- Delayed decisions — no live tracking
- Manual traceability — error-prone
- Overstock in one plant, shortage in another

✓ AFTER

Plant 1 ✓
Plant 2 ✓

MES + ERP
Central Platform

Plant 3 ✓
Plant 4 ✓

All plants connected in real-time

- **Centralized visibility — all 5 plants unified**
- **Standardized SOPs, quality checks & reporting**
- **Real-time dashboards — instant decisions**
- **Automated end-to-end batch traceability**
- **Balanced inventory with inter-plant transfers**

Data	Visibility	Quality	Traceability	Planning
Siloed → Centralized	Delayed → Real-time	Inconsistent → Standardized	Manual → Automated	Reactive → Predictive

KEY SUCCESS FACTORS

01

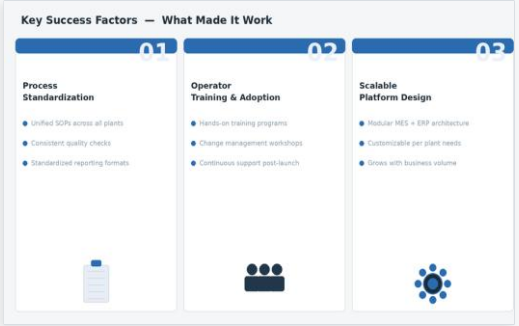
Process standardization before digitization

02

Operator training and adoption

03

Scalable and customizable software platform



LESSONS LEARNED

Technology alone is not enough	Process alignment is critical
Start with a pilot plant	To reduce risk before full rollout
Data accuracy is the backbone	Of successful plant unification
Change management is key	For operator and manager adoption

The unified platform ensures

5 plants, 1 system.

Centralized control. Decentralized execution.

30%↓	25%↑	IATF 16949
Defects	Visibility	Compliant